

DESIGN-A-THON RVCE 2026

DESIGN-A-THON



Launch : 1st June, 2026



Design-A-Thon Final Contest: 13th June 2026

Honeywell

RVCE

DP1: Sreedhar Garbham / Ibrahim Mohideen

DP2 : V S, Prakash

DP3: Anoop Moodambail

DP4: Siva Bavisetti / Ravi Thade

DP5: Jayakumar / Yogesha

DP6: Srivastava, Romi

DP7: Faiz / Renju

DP8: Tanmay/Venkatramani

General: Venkat Ganji, Ibrahim Mohideen

DP1:

DP2:

DP3:

DP4:

DP5:

DP6:

DP7:

DP8:

General: Prof Uttara Kumari



Rules :

- Each group/team can have max of 4 members
- A team can choose on any one problem only. Multiple teams can work on the same problem.
- Any software or open sources can be used, but needs to be referenced/acknowledged
- Complete solution along with the journey needs to be presented during the event
- Tips to a better score: Problem Understanding; Planning/Approach; Innovativeness of the solution; Presentation



Prizes :

- Winning team will get an opportunity for a Student Research Project (Duration TBD)
- Winner Prize Award Money: Rs.75000
- Runner-up Prize Award money: Rs.50000
- Third Prize (2) Award money: Rs. 25000 * Two third Prizes

DESIGN-A-THON

1st June : Launch/Kickoff

SME Session; Guidelines

06th June : Pulse Check/Connect

Q&A, Guidance

08th June: Submission Closes

10th June : Shortlisting of Solutions

Reviews by SMEs

12th June: Design-A-Thon

In person 3 hour event

DP1: TERRAIN GUARD - TERRAIN COMPLEXITY ASSESSMENT TOOL

Target Departments: CSE, ECE, IT, Aerospace

Goal: Terrain-related accidents (CFIT) account for ~25 % of all fatal aviation accidents globally. Existing terrain databases are static and do not provide a unified complexity score that can be embedded in real-time avionics, flight planning tools and preflight risk assessment indicator. The goal is to derive Terrain Topography Complexity Index (TTCI) from open-source DEM data and build a safety application.

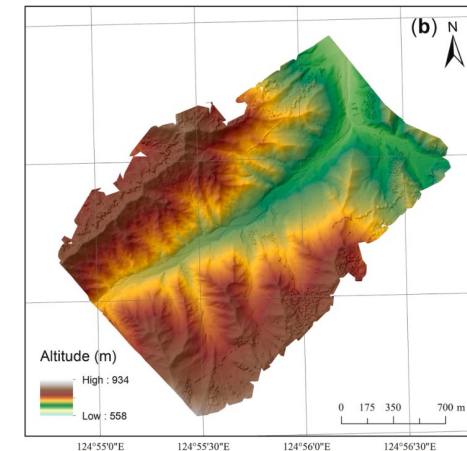
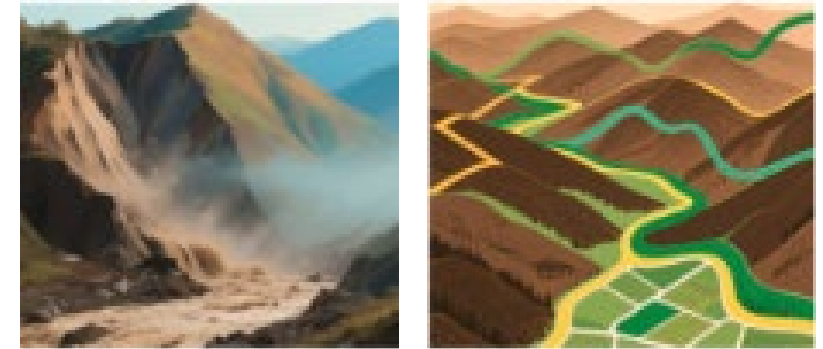
Open-Source Datasets:

1. [SRTM 30 m \(NASA / USGS\)](#)
2. [Copernicus GLO-30 DEM \(ESA\)](#)
3. [OpenTopography \(multi-source API\)](#)

Deliverables:

- TTCI computation pipeline (Python / GDAL / rasterio script)
- Interactive TTCI risk map (Leaflet / QGIS plugin)
- Aviation safety feature (choose one):
 - TAWS-style alert overlay
 - MSA calculator per route sector
 - UAS corridor risk scorer

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<https://www.sciencedirect.com/science/article/pii/S2666017225000719>

DP2: AIRFOIL SHAPE OPTIMIZATION

Maximize Aerodynamic Efficiency Using the UIUC Airfoil Coordinates Database

Target Departments: Mechanical, Aerospace, ECE

THE CHALLENGE

Using the UIUC Airfoil Coordinates Database (~1,650 real-world airfoil profiles), select a baseline airfoil and optimize its shape to achieve the highest lift-to-drag ratio (L/D) under the given flight conditions and geometric constraints.

You may modify existing coordinates or generate entirely new profiles. Any analytical, computational, or data-driven method is permitted.

THE DATASET

UIUC Airfoil Coordinates Database
University of Illinois at Urbana-Champaign

- ▶ ~1,650 airfoil profiles in .dat format
- ▶ Each file: airfoil name + (x, y) coordinates
- ▶ Families: NACA, Eppler, Wortmann, Selig, Clark-Y, Göttingen, ONERA & more

Download (ZIP — all airfoils in one file):
m-selig.ae.illinois.edu/ads/archives/coord_seligFmt.zip

DELIVERABLES & JUDGING CRITERIA

- ① Final airfoil coordinates (.dat file) (Have extracted values)
- ② Design rationale & methodology report
- ③ Performance comparison: baseline vs optimized
- ④ Visualization of pressure / velocity fields (bonus)

(Max 5 Slides)

Judging Weights:

- ▶ Max L/D at $Re = 500,000$, $AoA = 5^\circ$ — 35%
- ▶ Innovation in approach — 25%
- ▶ Structural feasibility (min $t/c \geq 10\%$) — 20%
- ▶ Presentation & documentation — 20%

DESIGN CONDITIONS

- ▶ Reynolds Number: 500,000 ▶ Mach: 0.1 (low-speed, incompressible)
- ▶ Angle of Attack: 0° to 10° (optimize for peak L/D)
- ▶ Min thickness-to-chord: 10%

MANDATORY RESULTS TO REPORT

Your submission MUST include these values:

- ▶ Baseline airfoil name (e.g. NACA 2412)
- ▶ Baseline L/D, Cl , Cd at $AoA = 5^\circ$, $Re = 500k$
- ▶ Optimized airfoil .dat file (x, y coordinates)
- ▶ Optimized L/D, Cl , Cd at $AoA = 5^\circ$, $Re = 500k$
- ▶ Peak L/D and the AoA at which it occurs
- ▶ Max thickness-to-chord ratio (t/c %)
- ▶ Cl vs AoA polar plot (0° to 10°)
- ▶ Cd vs AoA polar plot (0° to 10°)
- ▶ % improvement in L/D over baseline
- ▶ Tool / method used (XFOIL, CFD, ML, etc.)

ALLOWED APPROACHES

- ✓ Parametric (NACA / CST / PARSEC) shape optimization
- ✓ **ML / Deep Learning surrogate models (Preferred)**
- ✓ Genetic Algorithm / PSO / Bayesian optimization
- ✓ Panel method / CFD-coupled design loops

DP3: BOLT LOAD DISTRIBUTION IN A FLANGED JOINT

Target Departments: Mechanical, Aerospace

Bolted flange joints are widely used in structural and mechanical systems to connect cylindrical components. Accurate estimation of bolt forces under combined loading is essential for safe and efficient design. While finite element (FE) analysis provides detailed insight, classical hand calculations are often used for quick evaluation.

The **objective** of this problem is to determine the axial load in each bolt of a bolted flange joint subjected to combined loading using:

- Analytical (hand calculation) methods
- Finite Element (FE) analysis

Implement a 1D beam element model to represent the bolts in the FE analysis

Compare the results from both approaches and evaluate their agreement.

Identify the bolt experiencing the maximum load

Dimensions:

- Two concentric cylinders with 40-inches diameter; Cylinder thickness: 0.5 inches; length of each cylinder: 30 inches
- 8 standard bolts and nuts around the circumference equally spaced
- Flange height ~1.5 inches; Flange thickness 0.5 inches
- Bolt diameter ~0.25 inches (model nuts and bolts with approximate dimensions); bolt preload: 1500 lbf;

Materials: Steel (all components)

Boundary conditions:

- Fixed one end of the cylinder in all directions
- Apply $F_x=5000$ lbf axial force & bending moment of $M_y=100,000$ in-lbf on the free end of the second cylinder

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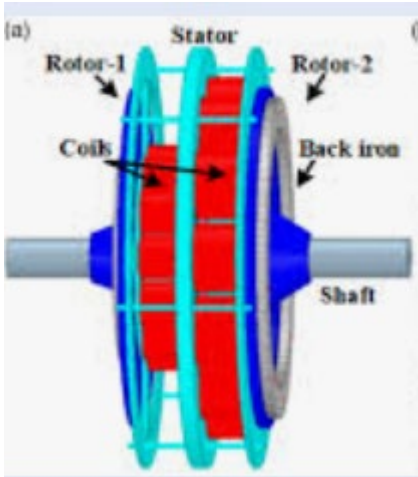


DP4: DESIGN OF AXIAL-FLUX PERMANENT-MAGNET LOW-SPEED MACHINES (GENERATOR)

Target Departments: Mechanical, Aerospace, EEE, ECE

Problem Description: Design of axial-flux permanent-magnet low-speed machines and performance comparisonFor Reference only between radial-flux and axial-flux machines comparison- (Only Analytical model)

Parameter	Requirement
Rated power	1 KW
Speed	400-600 RPM
Rated phase voltage	220
Rated torque	25-40NM
Efficiency	>90%
Power Factor (PF)	0.9
Weight	<5Kg



2. Final Report: Compile a comprehensive report summarizing the design outcomes with Size, Weight and Performance of Axial Flux generator and trade-off studies

Mentor:

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Electrical machines, Mechatronics, Smart Mobility, Mechanical design

DP5: GROUPING FUNCTIONALLY RELATED REQUIREMENTS

Target Departments: CSE, IT, ECE

Goal: Given set of requirements, the AI solution must group logically related textual requirements, build dependency tree, provide rationale for grouping.

Logical grouping: Organize textual requirements into clusters based on functional similarity.

Dependency mapping: Identify hierarchical and sequential relationships among requirements based on inputs, pre-conditions & outputs for each requirement.

Solution:

- On-prem open source Large Language Model (preferably Qwen 3.5 or equivalent) scalable to handle 500 reqs with 150+ pages
- No manual intervention till all requirements are analyzed
- API to be provided for users

Deliverables

- A categorized list of requirements grouped by logical similarity (e.g., logically related, performance).
- Dependency tree diagram: A visual or textual representation showing how requirements depend on each other.
- Rationale document: Explanation of why each requirement belongs to a specific group, including dependency justification.

Mentor:

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DP6: ACCENT SHIFT – EMOTION-PRESERVING ACCENT CONVERSION

Target Departments: CSE, ECE, IT

Problem Statement:

Design and build an end-to-end speech processing pipeline that takes an English speech audio file as input and outputs the same speech in a different English accent, while preserving the original speaker's emotion, prosody, and voice intensity.

Input

A single audio file containing spoken English

The input speech may have:

- Different English accents (e.g., American, British/London, Mexican English)
- Multiple emotions (e.g., happy, angry, sad, neutral)
- Variation in pitch, energy, and speaking rate

Output

An audio file where:

- The accent is converted to a target accent chosen by the user (e.g., Chinese-accented English, Japanese-accented English, Indian English, etc.)
- The speaker identity is not preserved (speaker-agnostic)
- The emotion, expressiveness, and voice intensity from the original audio remain perceptually intact
- Extend the solution to work real time when hosted on edge computer.

Evaluation Criteria:

- Accent accuracy (how well the target accent is achieved)
- Emotion preservation (perceptual similarity of emotional tone)
- Voice naturalness and fluency
- Scalability to multiple accents
- Innovation in model design or pipeline architecture

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DP7: AUTONOMOUS NAVIGATOR FOR GPS-DENIED ENVIRONMENTS

Target Departments: CSE, ECE, IT

Goal: To develop a Visual Inertial Odometry (VIO) system that enables accurate self-localization of a simulated drone in a GNSS - denied indoor environment.

Design Considerations:

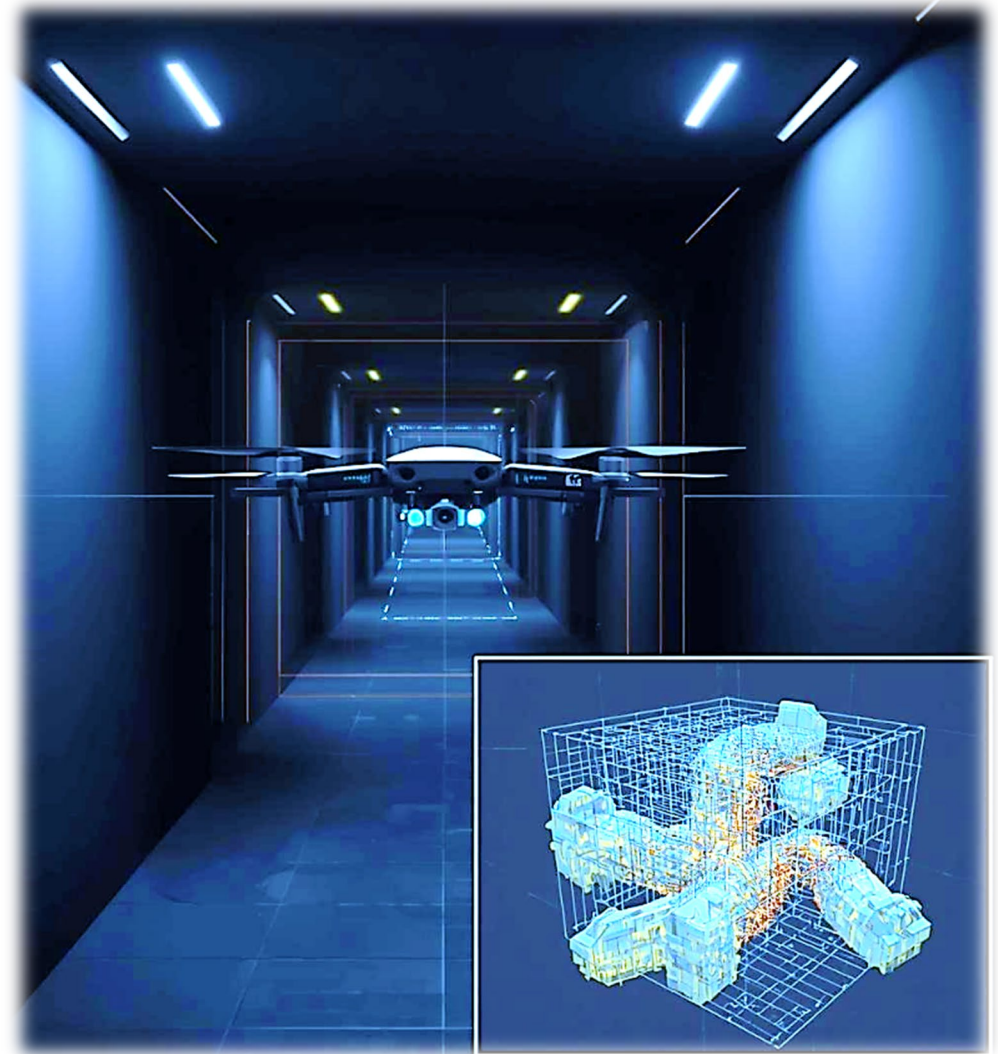
1. State Estimation: Visual-Inertial Odometry (VIO) drift < 1.5% over 200m.
2. Environmental: simulated indoor space (tunnel/warehouse) without GPS
3. Framework: ROS2 based pipeline running gazebo or Airsim.

Deliverables:

1. Simulation Setup: A Gazebo/Airsim environment replicating a tunnel/warehouse with a simulated drone equipped with a stereo camera and IMU.
2. VIO Pipeline :A working ROS2 node performing a real time visual-inertial odometry on the simulated sensor data.
3. Performance Report: Comparative graphs showing *Estimated Trajectory* vs *Ground Truth* (extracted from the simulator).
4. Source Code: Documented ROS2 workspace including custom VIO node and launch files.

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DP8: DESIGN AND DEVELOPMENT OF A VENTURI-BASED VACUUM GENERATION SYSTEM USING COMPRESSED AIR

Target Departments: Mechanical, Aerospace

Problem Statement:

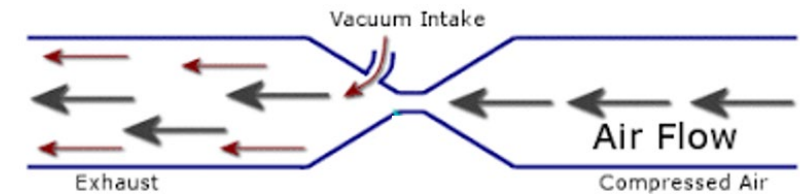
Design a Venturi-based system using a compressed air supply (9 bar, 100 LPM) to generate a vacuum sufficient to induce external air suction through an auxiliary line, achieving a target vacuum level of 0.14 to 0.54 bar absolute.

Objective

To develop and optimize a Venturi geometry that efficiently converts the given compressed air input into the required vacuum level while maintaining stable suction performance.

Deliverables

- Conceptual and detailed design of the Venturi device
- Calculations and assumptions used for design
- Performance analysis (pressure, flow, vacuum achieved) using Mathematical/CFD
- CAD model and brief design report



Concept of venturi effect

Evaluation Criteria:

- Understanding of Principles – Clarity in applying fluid dynamics and Venturi effect concepts
- Design Approach – Logic, assumptions, and methodology used in developing the Venturi
- Calculations & Accuracy – Correctness and completeness of pressure, flow, and vacuum calculations
- Performance Achievement – Ability of the design to meet the target vacuum (0.14 to 0.54 barA)
- Innovation & Optimization – Efficiency, creativity, and improvements in design
- CAD Model & Analysis – Quality of modeling and accuracy of simulation (mathematical/CFD)
- Report Quality – Conciseness, structure, and documentation of results

Mentor:

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SUPPORTING SLIDES

YOUR DATASET — UIUC AIRFOIL COORDINATES DATABASE

One dataset. ~1,650 airfoils. Everything you need to start optimising.

ABOUT THE DATASET

The UIUC Airfoil Coordinates Database is maintained by Prof. Michael Selig at the University of Illinois at Urbana-Champaign. It is the world's largest open-access collection of airfoil coordinate data.

- ▶ ~1,650 airfoil profiles in standard .dat format
- ▶ Each file: airfoil name (line 1) + x, y coordinates
- ▶ Coordinates: upper-surface TE → leading edge → lower-surface TE (counter-clockwise order)
- ▶ Families: NACA, Eppler, Wortmann, Selig, Clark-Y, Göttingen, ONERA, Drela, Liebeck, ARA & more
- ▶ Compatible with XFOIL, OpenFOAM, SU2, Python, MATLAB

HOW TO DOWNLOAD

- 1 Go to the UIUC Airfoil Data Site**
https://m-selig.ae.illinois.edu/ads/coord_database.html
- 2 Download the complete ZIP archive**
https://m-selig.ae.illinois.edu/ads/archives/coord_seligFmt.zip
- 3 Extract — you get ~1,650 individual .dat files**
Each file is a separate airfoil ready for analysis
- 4 Pick a baseline airfoil & start optimising!**
Suggested starters: NACA 2412, Clark-Y, E387, S1223

SAMPLE .DAT FILE FORMAT

NACA 2412	← Airfoil name
1.000000 0.001300	← x, y (upper surface TE)
0.950000 0.011400	← coordinates continue...
...	
0.000000 0.000000	← leading edge
...	
1.000000 -0.001300	← lower surface TE

KEY FACTS

- ▶ Free & open-access — no registration or login needed
- ▶ No licensing restrictions for academic / educational use
- ▶ Last updated: February 2026
- ▶ Maintained by Prof. Michael Selig, Dept. of Aerospace Eng.
- ▶ Files work directly with XFOIL (just drag & drop)